Felipe **Cybis Pereira**

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Research

PhD, neuroscience, spatial navigation and functional ultrasound imaging — Paris, France

@ Physics for Medicine Paris, Inserm, ESPCI Paris, PSL, CNRS — 2021-2025

- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Title: Revealing vascular patterns in the spatial navigation system with functional ultrasound imaging.
- Techniques: **Functional Ultrasound imaging - Spatial navigation - Rodent's brain imaging - Animal behavior and cognition - Python - 3D design conception (CAD) - Ultrafast Ultrasound**
- Recognition: PariSanté Campus PhD Award 2025 (Innovation & Industry)

Research Intern, neurobiology and ethology — Boston, USA

@ Rogulja Lab, Harvard Medical School, Harvard University — 2019-2019

- Adviser: Alexandra Vaccaro, PhD, and Dragana Rogulja, PI.
- Focus: Insights in sleep deprivation in *Drosophila melanogaster*.
- Techniques: ***Drosophila melanogaster*** **rearing - Immunostaining - Confocal microscopy - Ethological analysis - Survival and Dietary assays - Drosophila Activity Monitoring (DAM) system**

R&D Intern, functional ultrasound imaging — Paris, France

@ ICONEUS, real-time portable functional ultrasound small animal neuroimaging — 2018-2018

- Adviser: Bruno Osmanski, PhD, and Mickaël Tanter, PI.
- Focus: Code optimization for transcranial multiplane wave ultrasound imaging.
- Techniques: **Plane wave and Multiplane wave ultrasound imaging - Power Doppler imaging - Brain connectivity - MATLAB - Small animal brain imaging**

Teaching

Teacher assistant: Practical work in physiology and neuroscience — Paris, France

@ ESPCI Paris - PSL University — 2022-2024

- Professor: Thierry Gallopin, PhD.
- Neuroscience-focused practical work: **(1) human EEG (2) pose-estimation and animal tracking (3) human sleep**.

Teacher assistant: Methods for measuring and actuating neuronal activity — Saclay, France

@ M2 Computational Neurosciences and Neuroengineering, NeuroPSI — 2023-2025

- Professor: Isabelle Ferezou, PhD.
- Principles on Ultrafast Ultrasound, functional Ultrasound imaging and Ultrasound Localization Microscopy.

Education

BioMedical Engineering Master — Paris, France

Master's Degree in Bioengineering and Neurosciences — 2020-2021

- Scholarship student for the PSL Graduate Program in Life Sciences.

ESPCI Paris - PSL University — Paris, France

Master's Degree in Engineering focused in Biotechnology — 2016-2019

- *Michelin Excellency* scholarship student in a double degree program with UFSC University in Brazil.

Universidade Federal de Santa Catarina (UFSC) — Florianópolis, Brazil

Bachelor's Degree in Chemical Engineering — 2014-2020

Skills

Programming

- **Scientific Python**: Neuroimaging (Nipy suite, BrainGlobe), Machine learning (scikit-learn), Visualization (VTK, PyVista).
- **General Python Packaging**: Documentation and examples, unit testing, linting and formatting, pre-commit hooks.
- **Version control**: Intermediate to advanced usage of **Git**, **GitHub** and **GitHub Actions**.
- **Prototyping**: Intermediate knowledge in **Arduino** and **Raspberry Pi**.
- Intermediate knowledge on **Rust** (and **Rust bindings for Python**), **Lua**, **core-utils** and **shell-scripting**, **JavaScript**.
- Intermediate knowledge on text editing softwares such as LaTeX and **Typst**.

Spoken languages: **English** (fluent), **French** (fluent), **Portuguese** (native), Spanish (beginner)

Hobbies

I like playing **Tennis** (I played a lot while kid/teen), programming and tweaking my own dotfiles.

Publications and preprints

Cybis Pereira, F., Castedo, S. H., Meur-Diebolt, S. L., Ialy-Radio, N., Bhattacharya, S., Ferrier, J., Osmanski, B. F., Cocco, S., Monasson, R., Pezet, S., & Tanter, M. (2026). A Vascular Code for Speed in the Spatial Navigation System. *Cell Reports*, *45*(1). https://doi.org/10.1016/j.celrep.2025.116791

Le Meur-Diebolt, S., **Cybis Pereira, F.**, Mariani, J.-C., Kliewer, A., Farinha-Ferreira, M., Bertolo, A., Osmanski, B.-F., Lenkei, Z., & Deffieux, T. (2026). Robust Functional Ultrasound Imaging in the Awake and Behaving Brain: A Systematic Framework for Motion Artifact Removal. *Imaging Neuroscience*. https://doi.org/10.1162/IMAG.a.1191

Zucker, N., Le Meur-Diebolt, S., **Cybis Pereira, F.**, Baranger, J., Hurvitz, I., Demené, C., Osmanski, B.-F., Ialy-Radio, N., Biran, V., Baud, O., Pezet, S., Deffieux, T., & Tanter, M. (2025). Physio-fUS: A Tissue-Motion Based Method for Heart and Breathing Rate Assessment in Neurofunctional Ultrasound Imaging. *Ebiomedicine*, *112*, 105581. https://doi.org/10.1016/j.ebiom.2025.105581

International conferences

Bourgeais Rambur, L., Travert, Y., **Cybis Pereira, F.**, Huang, J., Baudouin, C., Melik-Parsadaniantz, S., Deffieux, T., Pezet, S., & Réaux-Le-Goazigo, A. (2023). Ultrafast Ultrasound Imaging of the Trigeminal Ganglion and Brain in Mice. *2023 ARVO Annual Meeting*, *64*, 3361.

Cybis Pereira, F., Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., & Tanter, M. (2022a,). Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. *Neuroscience 2022*.

Cybis Pereira, F., Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., & Tanter, M. (2022b,). Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. *fUSbrain 2022*.

Huang, J., Maguelone, F., Vetere, G., Pezet, S., Travert-Jouanneau, Y., **Cybis Pereira, F.**, Melik-Parsadaniantz, S., Amar, L., Bourgeais Rambur, L., & Réaux-Le-Goazigo, A. (2024). Deciphering the Precise C-Fos Connectome of Ocular Pain in Mice. *2024 ARVO Annual Meeting*, *65*, 2637.

Pereira, F. C., Ialy-Radio, N., Bhattacharya, S., Nouhoum, M., Osmanski, B.-F., Pezet, S., & Tanter, M. (2024). Functional Ultrasound Tools for Automatic Atlas Registration and Chronic Neuroimaging on Naturally Behaving and Sleeping Rats. *Sleep Medicine*, *115*, S409. https://doi.org/10.1016/j.sleep.2023.11.1098